

- **SOO/Project/Program Title**

RTS Mission Voice IP Solution

- **Purpose**

The purpose of this project is to replace the legacy voice communication system with a modern, scalable, and secure system to support mission-critical communications across the Morrell Operations Center.

This acquisition plan outlines the strategy for procuring and implementing a Mission Voice Communication system, hardware, and associated professional services solution, leveraging professional-grade intercom and audio products. The new solution will enhance reliability, scalability, and interoperability while aligning with modern digital communication standards and cybersecurity requirements.

This requirement is for an RTS ODIN/RVON-based mission voice/intercom solution. This solution must meet all salient characteristics in Section 6 and must achieve required interoperability, virtualization/remote operations support, operator-driven net management, and phased legacy transition.

- **Background**

The existing voice communication system presents an increasing operational risk due to degraded reliability. It is used across multiple mission-critical functions, including:

- Party nets for mission team coordination
- Point-to-point operator/lead conversations
- Secure conversations during launch operations and system checks

This Mission Voice IP Solution shall offer a modern, IP-based intercom solution that can integrate with current infrastructure while enabling a path forward to network-centric communications, improved voice quality, and centralized management.

- **Scope of Work**

The contractor shall design, deliver, and support installation of the Mission Voice IP-based intercom system that supports secure, real-time voice communication across designated facilities and meets Department of War (DoW) cybersecurity requirements.

The contractor shall support a phased transition from legacy voice interfaces and operational workflows to the new IP-based capability while maintaining continuity of operations throughout migration. The solution shall be delivered as a vendor-supported capability and shall not require Government-developed custom software, custom gateway development, or one-off engineering to meet interoperability, virtualization, or transition requirements (Turn-Key).

The contractor shall provide comprehensive, role-based training and knowledge transfer to ensure the Government's range support contractor personnel and system operators can independently operate, administer, and configure the Mission Voice system for future missions without reliance on contractor unique expertise for routine mission execution.

The contractor shall develop and deliver a training program consisting of, at a minimum: (1) system administrator/configuration training for designated Range Contractor personnel responsible for mission voice configuration and sustainment, and (2) operator training for mission personnel who will use the system during real-time operations. Training shall be tailored to the Government's operational workflows and shall include instruction, guided hands-on labs, and scenario-based training events that replicate mission timelines (e.g., net/party-line setup, point-to-point communications, coordination circuits, configuration changes during countdown, and restoration actions following disruptions).

The contractor shall provide administrator/configuration training sufficient for Range Contractor personnel to independently perform, at a minimum: creation and modification of users, roles, and permissions; creation/modification of channels, party lines, and point-to-point circuits; endpoint/keypanel/soft-panel provisioning; profile creation and management; routine configuration changes required for disparate mission sets; management of interoperability/bridging interfaces; execution of startup/shutdown procedures; backup/restore; patch/upgrade procedures (as allowed by the Government); fault isolation and troubleshooting; and restoration of services following common failure modes.

The contractor shall provide operator training sufficient for personnel to independently perform mission operations using the delivered endpoints (hardware keypanels and approved soft clients), including talk/listen selection, channel monitoring, private line operation, rapid switching between nets, use of user profiles, and execution of contingency procedures (e.g., degraded network conditions, endpoint failures, or patch changes) necessary to sustain continuous mission communications.

The contractor shall provide training materials and references such as Government deliverables, including lesson plans, student guides, quick-reference job aids, and configuration playbooks/checklists for common mission scenarios. Materials shall be provided in an electronic format suitable for Government reuse and updates.

The contractor shall conduct a knowledge-transfer/turnover event after installation and prior to final acceptance consisting of: (1) supervised configuration of a representative mission event by Range Contractor personnel, and (2) a supervised operational run using trained operators. Final acceptance of the training effort shall include successful completion of these events with minimal contractor intervention, demonstrating that Range Contractor personnel can configure and support mission voice communications for future missions.

The contractor shall provide a post-training support period (e.g., remote/onsite as negotiated) to answer administrator and operator questions arising from initial operational use and to reinforce configuration procedures required for near-term missions.

- **Item List**

Item Description	Quantity
Section I	
DSO connections 600ea (320 SIP 320 RVON) Matrix Set	1
ODIN 128 port matrix	6
ODIN / RVON 64 ch expansion	7

ODIN 128 port matrix Redundant	1
ODIN Redundancy Management license	1
SFP Single Mode, 15 km, 2.67 Gbit/s	28
Software interface SIP - 8 ports - Software Product	
(Note: Server for SIP software not supplied)	40

Section II

Trunkmaster for bridging systems

TM-10K Gen 2 trunkmaster	2
TM-10K_GAIN SW gain control software	2

Section III

Control Room

ODIN 128 port matrix	2
ODIN / RVON 64 ch expansion	3
ODIN 128 port matrix Redundant	1
ODIN Redundancy Management license	1
SFP Single Mode, 15 km, 2.67 Gbit/s	12
VLINK Virtual App 8 users aka Handhelds	
RTS Vlink Software Intercom - 8 Port License -	
(Note: Server for VLINK app not supplied)	2

Section IV

End Instruments 0.80

KP5032 5F OPT-AC 32 position keypanels	25
PH-88R5 light weight single sided headset	25

• Performance Objectives

- Decommission the legacy voice system in a phased approach.
- Procure and deploy Mission Voice IP Hardware.
- Integrate any existing voice comm systems
- Ensure continuity of operations during migration

- Train users and administrators on the new Mission Voice IP platform
- Maintain compliance with DoW cybersecurity, interoperability, and sustainment standards
- Reduce voice communication outages.

- **Salient Characteristics**

- System Characteristics

- IP-based voice intercom leveraging an existing IP network backbone
- Characteristics specifically related to the desktop app
- Ability for individuals to drop voice channels into instrumentation
- Minimum 128-port rack-mounted voice matrix for mission-critical communications
- Accommodate a minimum of 600 end users
- Supports Session Initiation Protocol (SIP)
- Phased legacy transition: The solution shall support a low-risk, phased cutover from legacy voice environments and interfaces to IP operations while maintaining continuous mission communications.
- Legacy/IP bridging: The solution shall provide required interoperability between legacy interfaces and IP-based endpoints without government-funded custom gateway development or one-off engineering.
- Virtualization/remote ops: The solution shall support operation and administration in Government virtualized and remote operations environments (e.g., Government-hosted virtualization) with vendor-supported soft panels/virtual keypanels and management capability.
- Operator-driven net management: The solution shall support rapid, operator-level reconfiguration of party lines/nets and point-to-point communications during mission operations without requiring administrative reconfiguration for routine mission timeline changes.
- Console physical constraints: Console-attached keypanels/endpoints shall fit existing console physical constraints and shall not require console redesign, rework, or modification to install.
- External stakeholder interoperability: The solution shall support interoperability with external mission partners and Launch Service Provider (LSP) launch communications environments. Interoperability shall be achievable using vendor-supported capabilities and interfaces and shall not require replacing external stakeholder equipment as a condition of interoperation.
- Compatibility with legacy range system and interfaces (e.g., T1, Compunetix, Orion, MVP)
- Compatibility with external stakeholders who use different mission voice-com systems
- Redundant management of the footprint and connectivity across multiple geographically separated locations to include, but not limited to the Cape and Patrick
- Software management of channels/access (bridging system) – software management suites to allow voice protocols and user endpoints to include mobile/desktop/system-specific instruments
- Compatibility with remote administration access
- Software endpoints (virtual/soft-client/mobile), if provided, shall be RTS vendor-supported (e.g., RTS VLINK or equivalent RTS-supported capability), shall operate in the Government's virtualized/remote operations environments, and shall not require Government-developed custom software, unmanaged third-party drivers, or unsupported middleware. Console and soft-client audio interfaces shall support commonly available operator headsets/peripherals appropriate for mission

operations.

- Console Hardware Compatibility: Console-attached audio interfaces must provide support for industry-standard headset connectors, specifically the 5-pin (XLR-style) microphone headset standard that is commonly used in mission control, dispatch, and aerospace environments. This ensures that operators can reliably plug in both legacy and modern headsets without the need for additional adapters.
 - System controls/configures network-connected end instruments
 - Recording capability
 - Reliable and easy-to-use party-line functionality, where multiple users can talk and listen simultaneously on a shared channel.
 - Ensuring low latency and minimal audio dropouts
 - Ability to isolate specific audio sources or channels for monitoring or recording and to troubleshoot audio problems
 - Private Line/Point-to-Point for direct, confidential communication between two users.
 - Gracefully handle network disruptions, such as temporary outages or packet loss, to re-establish connections automatically and minimize audio dropouts.
 - Interoperability with Launch Service Providers (LSPs) Mission Communications
- User
 - Flexibility for the user to switch between end devices, including mobile, desktop, and network-connected instruments.
 - Operator configurable of the layout and selection of the voice channels
 - 25 hardware end instruments with user keypanel/endpoints and headsets
 - 8 soft client mobile app based compatible with common device operating systems (OS) (e.g., Apple, Microsoft, Android, Google, etc.)
 - Portability required across Network platforms devices (e.g., Server, Desktop, Mobile, Thin-client, etc.)
 - Portability required across Operating System platforms (e.g., Android, Apple, Linux, Microsoft, etc.)
 - Intuitive and customizable user interface experience for talk and listen capability on multiple voice channels
 - Allow users' profiles to save their preferred settings and be loaded quickly when switching devices
- **Place of Performance**
 - Cape Canaveral Space Force Station

Intentionally Blank